K3 Shelf Array Server

Native RISC-V Build Cluster Server

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SpacemiT K3 Shelf is available in N10 and N48 high-density rack server models based on the self-developed K3 AI CPU with the RVA23 Profile. Each node features an 8-core X100 RISC-V CPU at up to 2.4 GHz, delivering up to 60 TOPS.

Built on a native RISC-V architecture, it supports Koji and Open Build Service (OBS), with native compilation 10× faster than QEMU emulation on x86, significantly reducing OS, firmware, and open-source build times.

At just 15–25 W per node, it cuts power consumption by up to 60% versus x86 servers. Four 10GbE ports, integrated BMC management, remote serial console, and cloud debugging make it ideal for RISC-V software validation, source code build clusters, on-premises LLM deployment, and edge computing.

0/48 K3 compute nodes

Each K3 node features an 8-core X100 RISC-V CPU at up to 2.4 GHz and an 8-core A100 AI CPU, delivering up to 60 TOPS.

Mainstream Linux Build System Support

Supports Koji and Open Build Service (OBS), with native compilation over 10× faster than QEMU on x86.

RISC-V Software Compatibility Testing

The industry's first mass-produced RVA23 Profile array server for validating Linux distributions, open-source software, and AI frameworks.

Remote RISC-V Cloud Debugging

Supports remote flashing, serial console, SSH, Web IDE, and remote desktop for an out-of-the-box cloud development environment.

Exceptional Energy Efficiency

Twelve K3 nodes match the build performance of a mid-range x86 workstation while consuming just 15–25 W per node, cutting power costs by up to 60%.

Equipped with 4x 10GbE ports

Features 4×10GbE (SFP+) ports and a dedicated MGMT port for high-speed networking and BMC management.

Integrated BMC Management

Provides real-time monitoring, configuration, hardware management, diagnostics, firmware updates, and open APIs for software development.

Broad Application Scenarios

Ideal for AI computing, edge AI, on-premises LLM deployment, smart cities, healthcare, industrial AI, and intelligent security.

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Specifications - N10

Technical Specifications	Form Factor	1U rack-mounted AI computing server
	Architecture	RISC-V architecture
	Nodes	10 distributed compute nodes + 1 control node
	Compute Node	8-core 64-bit X100 processor, up to 2.4 GHz, with an 8-core AI CPU supporting 1024-bit RVV 1.0 vector computing
	Video Encoding	H.265/H.264: 1 × 4K@60fps, 8 × 1080p@30fps
	Video Decoding	4K@120fps [H.264/H.265/VP9]/JPEG(MPEG4/MPEG2), 16x1080P@30fps
	Control Node	8-core 64-bit X100 processor, up to 2.4 GHz, with an 8-core AI CPU supporting 1024-bit RVV 1.0 vector computing
	AI Performance	2880 TOPS (60 TOPS × 48, INT4)
	Memory	16 GB LPDDR5 × 10(16 GB / 32 GB options)
	Storage	128 GB UFS × 10 (16/32/64/128/256 GB options)
	Storage Expansion	Optional M.2 2280 PCIe NVMe SSD × 48 and one 3.5"/2.5" SATA 3.0 SSD with hot-swap support. The BMC provides direct storage management, while compute nodes can access the drive through BMC network sharing.
	Power Supply	550 W AC power supply (90–264 VAC, 47–63 Hz), non-hot-swappable
	Cooling	6 high-speed cooling fans
Physical Specifications	Dimensions	440.5 × 494.0 × 44.4 mm (L × W × H)
	Weight (Fully Configured)	Net weight: 8.1 kg; Shipping weight: 10.3 kg
	Environmental	Operating: 0°C - 35°C; Storage: -40°C - 60°C; Humidity: 5% - 80%RH (non-condensing)
Software	BMC	Integrated web-based BMC management system supporting monitoring, configuration, alarms, remote management, virtual media, CLI, and Redfish APIs for software development.
	LLMs	Supports on-premises deployment of full-parameter Transformer-based LLMs, including DeepSeek-R1, Gemma, Llama, ChatGLM, Qwen, Phi, and more.
	Deep Learning	Supports CNN, RNN, LSTM, and other neural networks, together with TensorFlow, PyTorch, PaddlePaddle, ONNX, Caffe, custom operators, and Docker containerization.
Interfaces	Networking	4 × 10GbE (SFP+) ports, 1 × Gigabit Ethernet (RJ45, MGMT for BMC)
	Console	1 × Console (RJ45, BMC debug port, 115200 bps)
	Display output	1 × HDMI (up to 1080p, BMC display output)
	USB	2 × USB 3.0 (supports USB OTG for BMC firmware upgrade)
	Buttons	1 × Reset, 1 × Power, 1 × BMC Reset
	Other Interfaces	1 × RS232 (DB9, 115200 bps), 1 × RS485 (DB9, 115200 bps)

Specifications - N48

Technical Specifications	Form Factor	2U rack-mounted AI computing server
	Architecture	RISC-V architecture
	Nodes	48 distributed compute nodes + 1 control node
	Compute Node	8-core 64-bit X100 processor, up to 2.4 GHz, with an 8-core AI CPU supporting 1024-bit RVV 1.0 vector computing
	Video Encoding	H.265&H.264: 1x4K@60fps, 8x1080P@30fps
	Video Decoding	4K@120fps (H.264/H.265/VP8/VP9/JPEG/MPEG4/MPEG2), 16x1080P@30fps (H.265& H.264)
	Control Node	8-core 64-bit X100 processor running at up to 2.4 GHz, paired with an 8-core A100 AI CPU delivering up to 60 TOPS
	AI Performance	2880TOPS（60TOPS × 48，INT4)
	Memory	16 GB LPDDR5 × 48 (16 GB / 32 GB options)
	Storage	128 GB UFS × 48 (16/32/64/128/256 GB options)
	Storage Expansion	Optional M.2 2280 PCIe NVMe SSD × 48
	Power Supply	Dual redundant AC power supplies (hot-swappable)
	Screen	1 × touchscreen display
	Cooling	12 high-speed cooling fans
Physical Specifications	Dimensions	724.0 × 430.0 × 88.8 mm (L × W × H)
	Weight (Fully Configured)	Net weight: 23.1 kg; Shipping weight: 25.3 kg
	Environmental	Operating: 0°C - 35°C; Storage: -40°C - 60°C; Humidity: 5% - 80%RH RH (non-condensing)
Software	BMC	Integrated web-based BMC management system supporting monitoring, configuration, alarms, remote management, virtual media, CLI, and Redfish APIs for software development.
	LLMs	Supports on-premises deployment of full-parameter Transformer-based LLMs, including DeepSeek-R1, Gemma, Llama, ChatGLM, Qwen, Phi, and more.
	Deep Learning	Supports CNN, RNN, LSTM, and other neural networks, together with TensorFlow, PyTorch, PaddlePaddle, ONNX, Caffe, custom operators, and Docker containerization.
Interfaces	Networking	2 × 10GbE (SFP+), 2 × Gigabit Ethernet (RJ45), and 1 × Gigabit Ethernet (RJ45, MGMT for BMC)
	Console	1 × Console (RJ45, BMC debug port, 115200 bps)
	Display output	1 × VGA (up to 1080p, BMC display output)
	USB	2 × USB 3.0 (supports USB OTG for BMC firmware upgrade)
	Buttons	1 × Reset, 1 × UID, 1 × Power button